Comparison of Early Versus Late Amniotomy with Immediate Oxytocin Infusion In Nulliparous women with bishop's score *leq*6 - A Randomised Trial

REGISTRATION NO: ???????????

DISSERTATION SUBMITTED TO

THE TAMIL NADU DR. MGR MEDICAL UNIVERSITY

IN PARTIAL FULFILMENT OF THE REQUIREMENTS

FOR THE AWARD OF THE DEGREE OF

M.S. OBSTETRICS AND GYNAECOLOGY

Branch II

SEPTEMBER 2026

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DECLARATION

In the following pages is presented a consolidated report of the study of "Comparison of Early Versus Late Amniotomy with Immediate Oxytocin Infusion In Nulliparous women with bishop's score $leq6$ - A Randomised Trial", on cases studied and followed up by me at Government Raja Mirasudhar Hospital, Thanjavur Medical College from 2025 - 2026. This thesis is submitted to the DR. MGR Medical University, Chennai in partial fulfillment of the rules and regulations for the award of MS Degree examination in Obstetrics and Gynaecology.

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Junior Resident,

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Thanjavur Medical College, Thanjavur

Tamil Nadu

CERTIFICATE BY THE GUIDE

This is to certify that this dissertation entitled "Comparison of Early Versus Late Amniotomy with Immediate Oxytocin Infusion In Nulliparous women with bishop's score $leq6$ - A Randomised Trial" is a bonafide research work done by M Saranya, under the guidance and supervision in the Department of Obstetrics and Gynaecology during the period of her postgraduate study for M.S. (Obstetrics and Gynaecology) from 2023 to 2026.

SIGNATURE OF THE GUIDE

Dr. P Amudha MDOG,

Professor and Head of Department,

Department of Obstetrics and Gynaecology

Government Raja Mirasudhar Hospital

Thanjavur Medical College,

Tamil Nadu

FILLER FOR PLAGIARISM ANALYSIS

<u>CERTIFICATE II</u>

This is to certify that this dissertation entitled "Comparison of Early Versus Late Amniotomy with Immediate Oxytocin Infusion In Nulliparous women with bishop's score *leq*6 - A Randomised Trial" of the candidate M Saranya, REGISTRATION NUMBER: ???????????for the award of MASTER OF SURGERY in the branch of Obstetrics and Gynaecology (BRANCH II). I personally verified the urkund.com website for the purpose of plagiarism check. I found that the uploaded thesis file contains from introduction to conclusion pages and result shows 2% of plagiarism in the dissertation.

GUIDE and SUPERVISOR sign with seal

ENDORSEMENT BY THE HOD

AND

HEAD OF THE INSTITUTION

This is to certify that this dissertation entitled“Comparison of Early Versus Late Amniotomy with Immediate Oxytocin Infusion In Nulliparous women with bishop's score $leq6$ - A Randomised Trial” is a bonafide research workd done by M Saranya, under the guidance and supervision of Dr. P Amudha MDOG Professor and Head of Department, Department of Obstetrics and Gynaecology, Goverment, Government Raja Mirasudhar Hospital Thanjavur Medical College, Tamil Nadu

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SIGNATURE OF THE DEAN

Dr. DEAN

Government Raja Mirasudhar Hospital,

Thanjavur Medical College,

Tamil Nadu

Contents

INTRODUCTION

Induction of labour is defined as the process of artificial initiation of uterine contraction, prior to spontaneous onset with intention to make progressive effacement and dilatation of cervix. Induction of labour is a commonly performed obstetric procedure.

AIM OF THE STUDY

PRIMARY OBJECTIVE:

To compare the time interval between oxytocin induction to delivery interval in women with early and late amniotomy

SECONDARY OBJECTIVE:

1. To compare the maternal and neonatal outcome in women with early and late amniotomy

JUSTIFICATION FOR THE STUDY

TBD

MATERIALS AND METHODS

SELECTION OF CASES:

This is a randomised controlled open label clinical intervention trial conducted at the Department of Obstetrics and Gynaecology, Government Raja Mirasudhar Hospital attached to Thanjavur Medical College from September 2024 to August 2025 including women admitted in the labour room who satisfy the inclusion and exclusion criteria. Patient identity will be kept confidential. Access to records for persons other than researchers and guide will be prohibited.

INCLUSION CRITERIA

- Age ¿18yrs
- Full term ¿37 weeks gestation
- Singleton pregnancy
- Fetus in cephalic presentation
- Bishop's Score$\geq$6
- Reassuring fetal heart rate.

EXCLUSION CRITERIA

- History of uterine surgery

- Previous cesarean section
- Ruptured membranes
- Macrosomia
- Gross congenital malformation,
- Estimated fetal weight¡3rdpercentile
- HIV
- Hepatitis B or C.

SAMPLE SIZE:

Sample size was calculated using openepi.com. From the previous study by Naskar A et al. 2022, the prevalence of first trimester bleeding was found to be 7.03%. Keeping 95% confidence interval, power as 80% and precision ass 5%, the sample size was calculated as 150.

STATISTICAL METHODS:

This prospective observational study is to be carried out on pregnant women attending OPD with history of first trimester vaginal bleeding at Govt.Rajah Mirasudar Hospital for a period of twelve months. Patients with history of amenorrhea and positive urine pregnancy test with bleeding per vaginum in first trimester are to be included. Complete history followed by complete examination (general, physical and gynaecological) of all the patients at initial visit to be taken. The patients are to be followed up regularly in antenatal clinic and repeat ultrasound scans to be done as required. A structured proforma will be used to collect information and followed till their pregnancy terminated. The vaginal bleeding histories includind the duration of bleeding in days, the severity of bleeding and associated abdominal pain are collected. Bleeding is to be categorized as spotting, moderate, or heavy bleeding according to the self-assessed degree of vaginal bleeding. BT, CT, PT/INR and APTT to be done of all the patients to rule out bleeding tendencies. Ultrasound examination will be done in all subjects who are included in this investigation. All such patients will be prospectively followed throughout the pregnancy and intrapartum period and their outcomes will be studied. Postpartum follow-up will be performed by telephonic talk or medical record review for 3 months. Data will be entered and analyzed using Microsoft excel and Descriptive analysis will be done.

REVIEW OF LITERATURE

review of literature

RESULTS

	MEAN(yrs)	**STD DEV(yrs)**	**p-value**
Early Amniotomy	25.91	5.45	0.9606
Late Amniotomy	25.96	4.77	

Table 1: Table showing the age distribution in the study population.

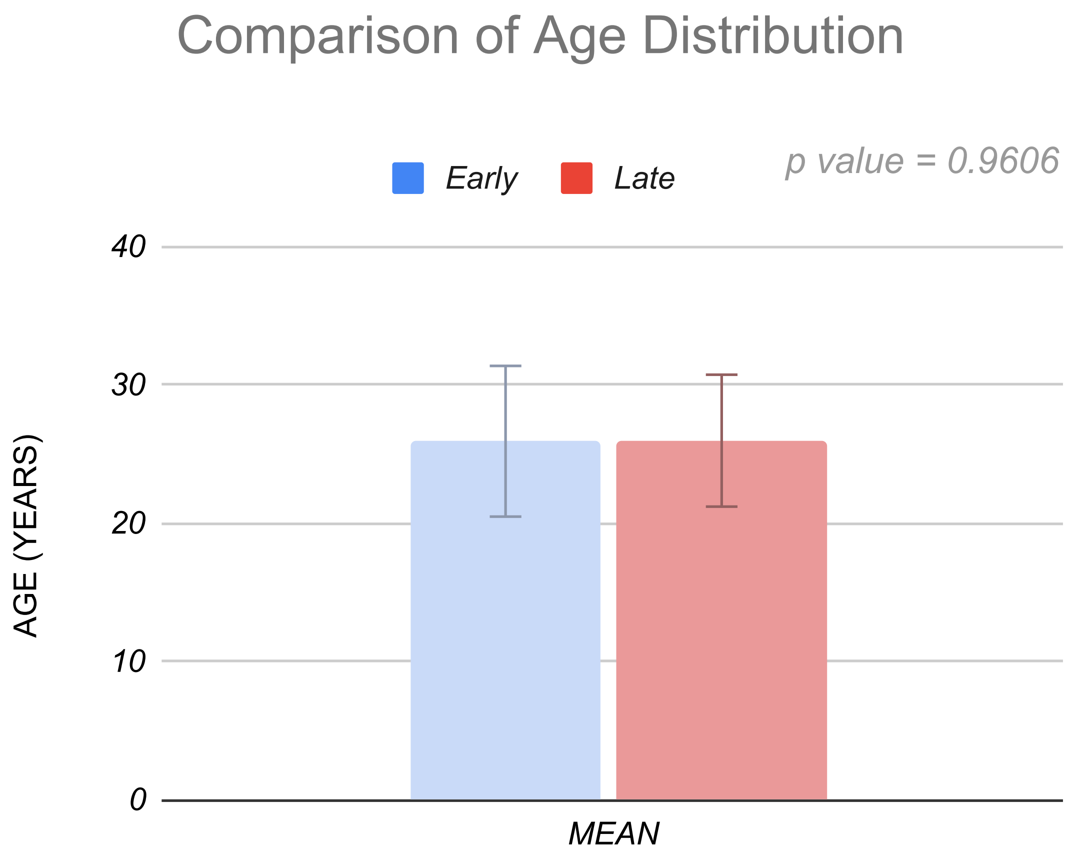

Figure 1: Chart showing the age distribution in the study population.

AGE GROUP	EARLY (n=70)	(%)	LATE (n=70)	(%)
<21	14	20	13	18.57
21-30	40	57.14	45	64.29
>30	16	22.86	12	17.14

Table 2: Table showing the distribution of women in different age groups in the study population

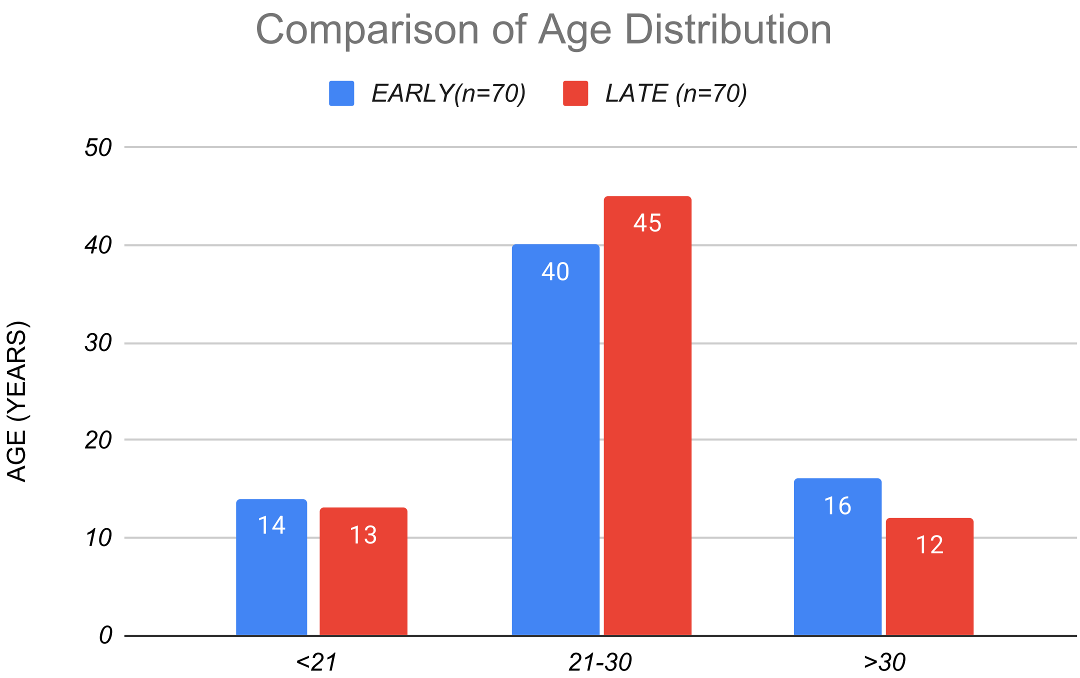

Figure 2: Chart showing the distribution of women in different age groups in the study population

	MEAN	**STD DEV**	**p-value**
Early Amniotomy	24.10	3.09	0.3531
Late Amniotomy	24.60	3.24	

Table 3: Table showing the BMI distribution in the study population.

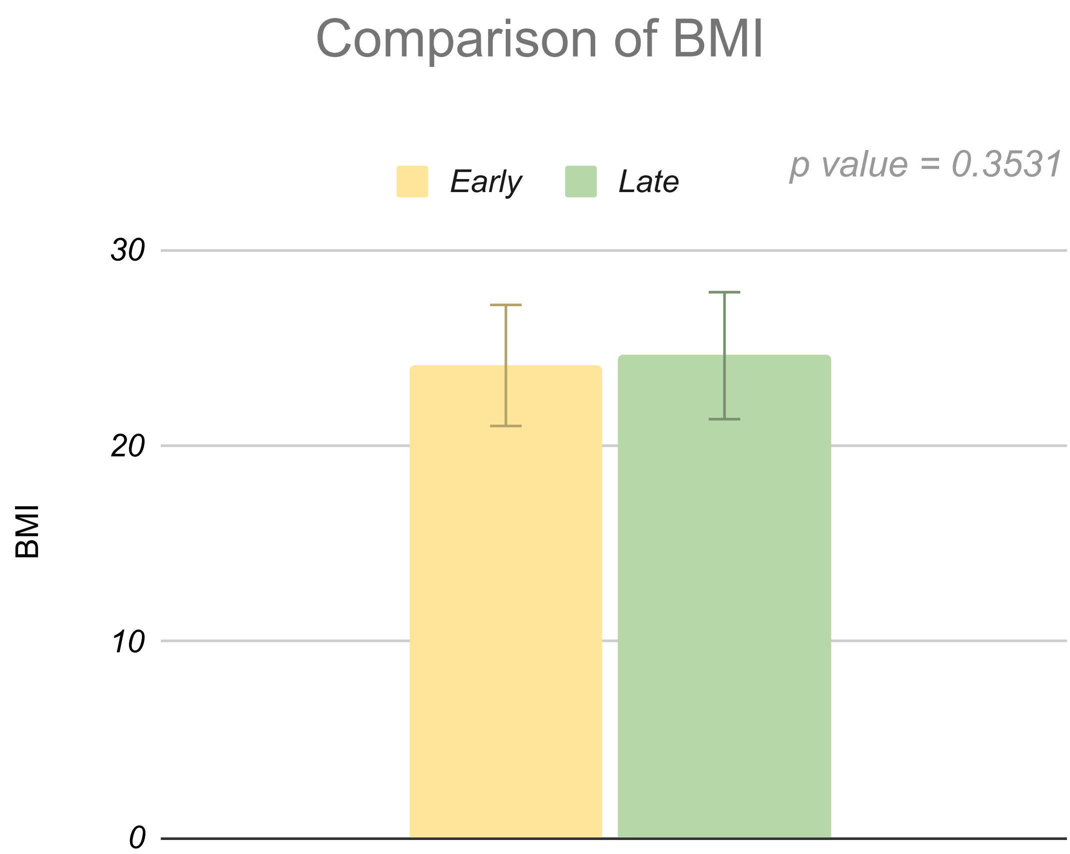

Figure 3: Chart showing the BMI distribution in the study population.

	MEAN(wks)	STD DEV(wks)	p-value
Early Amniotomy	38.61	1.17	0.3319
Late Amniotomy	38.43	1.08	

Table 4: Table showing the distribution of gestational age in the study population.

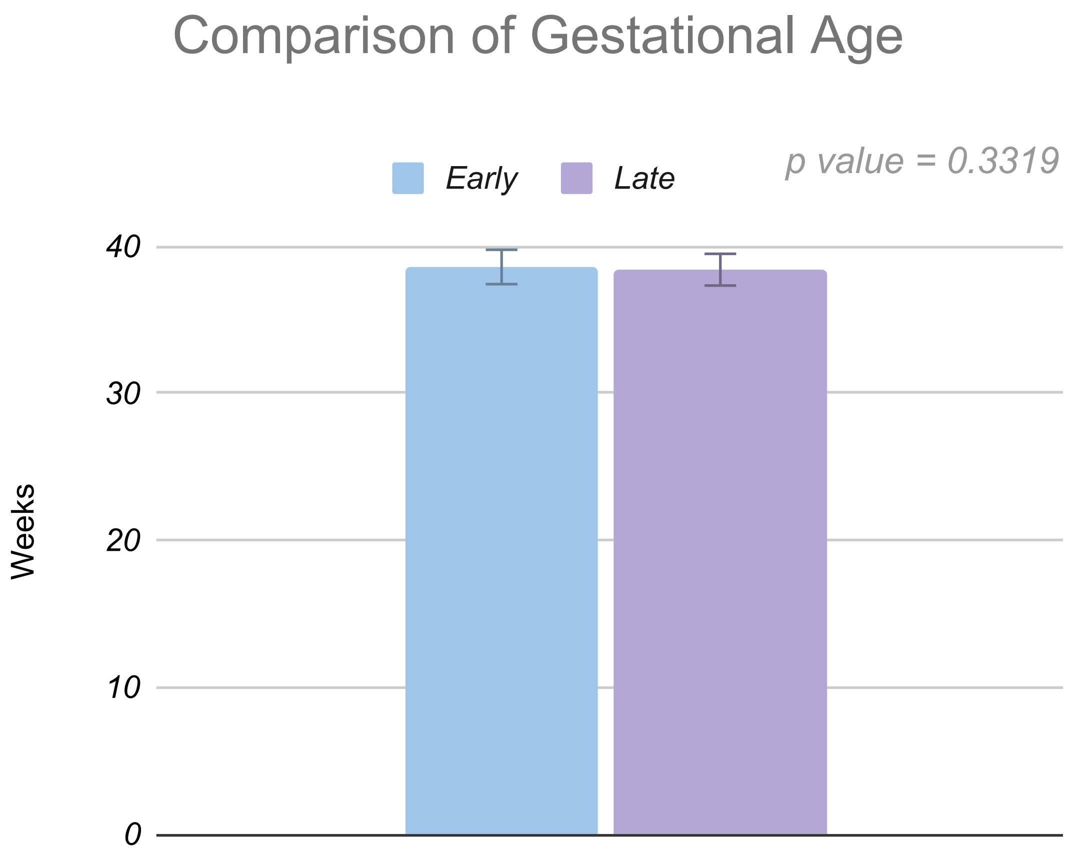

Figure 4: Chart showing the distribution of gestational age in the study population.

INDICATION FOR IOL	**EARLY (n=70)**	**(%)**	**LATE (n=70)**	**(%)**
GDM	22	31.43	25	35.71
Post dates	22	31.43	15	21.43
Hypertension	3	4.29	2	2.86
Preeclampsia	7	10	6	8.57
Oligohydramnios	16	22.86	22	31.43

Table 5: Table showing the various indication for induction of labour in the study population.

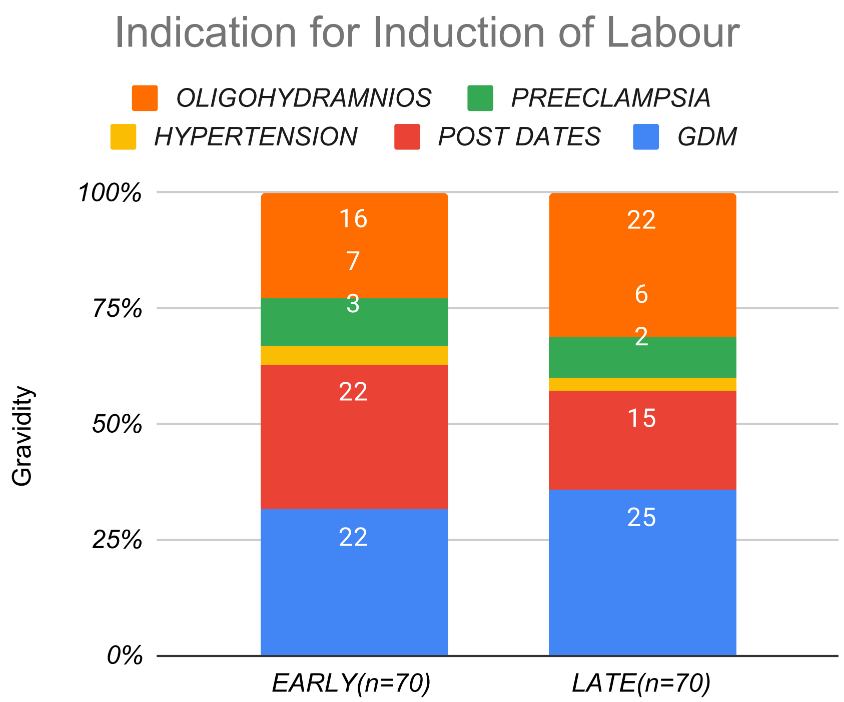

Figure 5: Chart showing the various indication for induction of labour in the study population.

	MEAN	STD DEV	p-value
Early Amniotomy	6.37	0.49	0.6067
Late Amniotomy	6.41	0.50	

Table 6: Table showing the distribution of bishop's score in the study population.

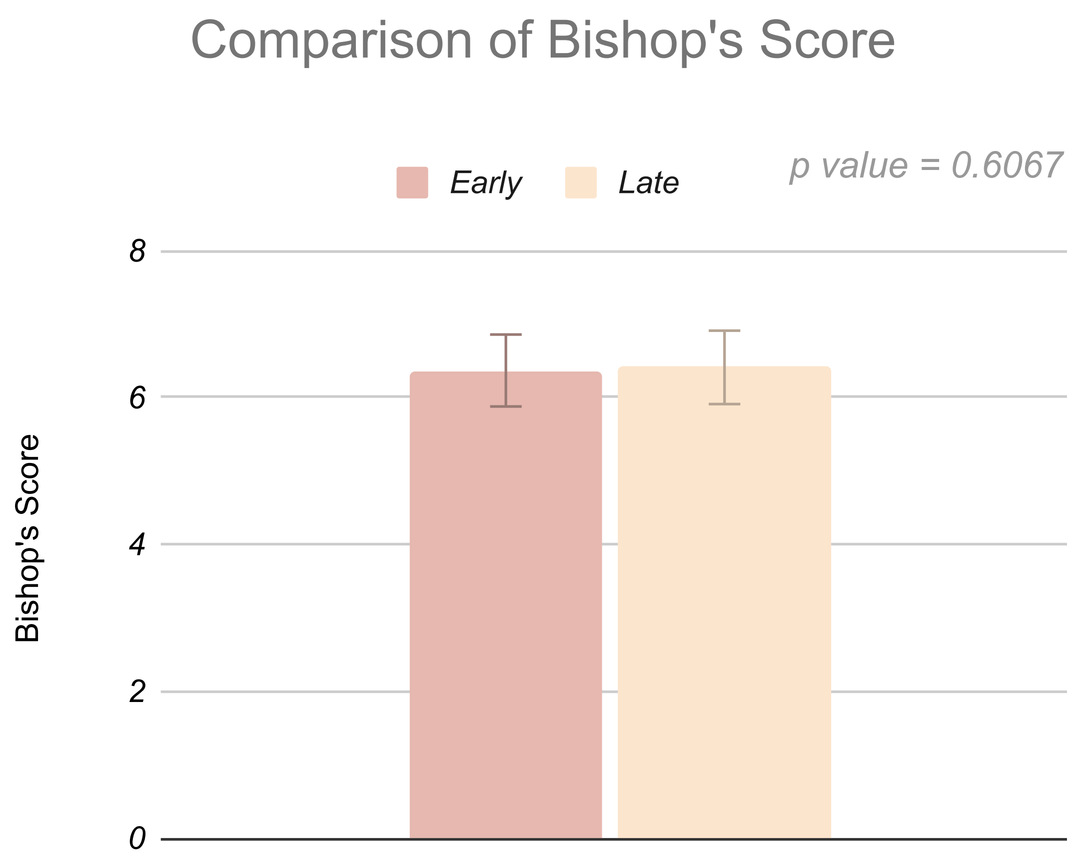

Figure 6: Chart showing the distribution of bishop's score in the study population.

	MEAN (hrs:min)	STD DEV (hrs:min)	p-value
Early Amniotomy	3:49	1:30	<0.0001
Late Amniotomy	7:58	2:57	

Table 7: Table showing the time taken to reach active phase in the study population.

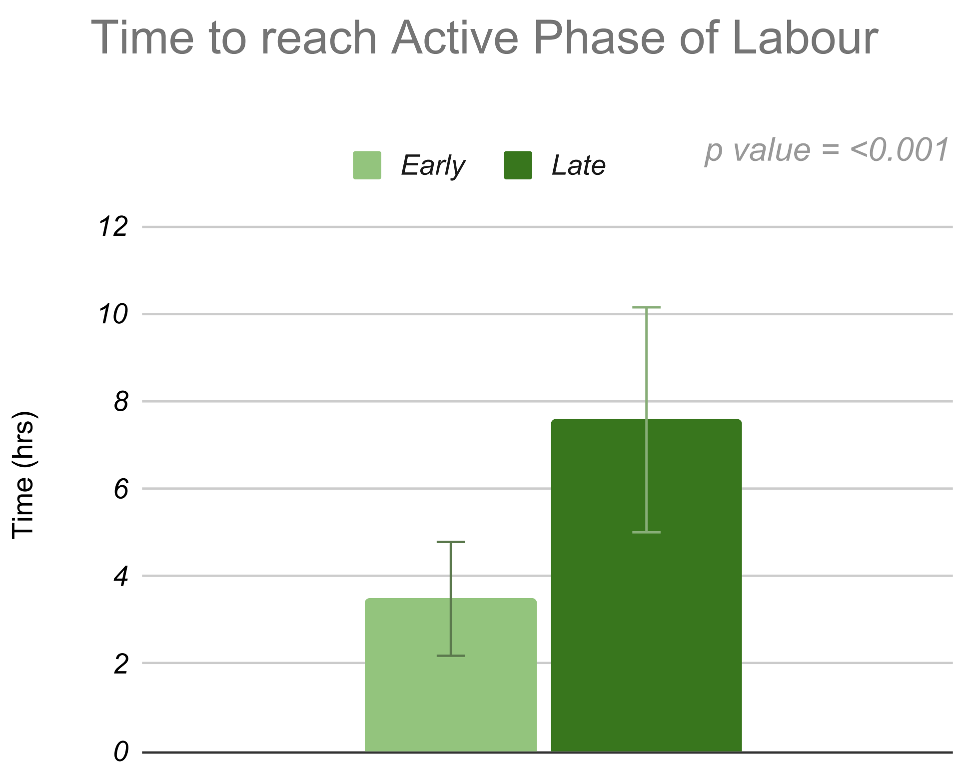

Figure 7: Chart showing the time taken to reach active phase in the study population.

	OXYOCIN INDUCTION TO DELIVERY(hrs:min)	**STD DEV (hrs:min)**	**p-value**
Early	2:16	1:28	0.2794
Late	1:59	1:20	

Table 8: Table showing the time interval between oxytocin induction and delivery in the study population.

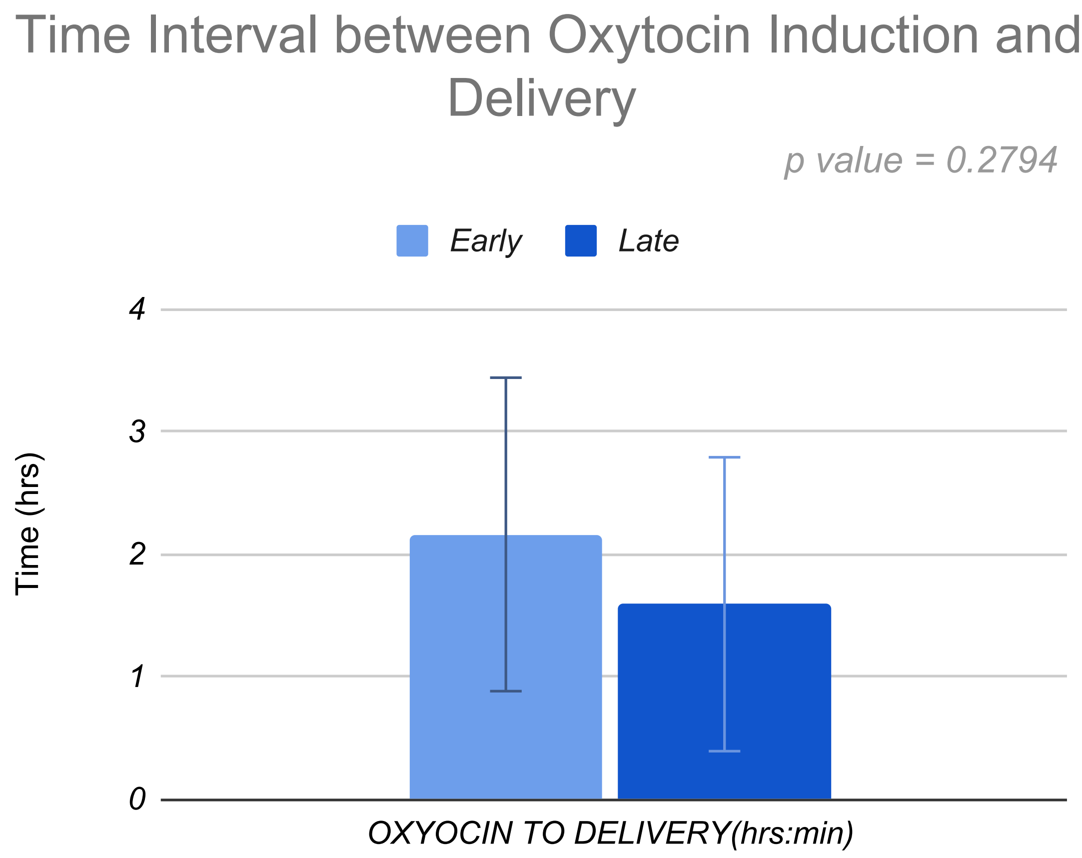

Figure 8: Chart showing the time interval between oxytocin induction and delivery in the study population.

	MEAN (hrs:min)	STD DEV (hrs:min)	p-value
Early Amniotomy	6:06	2:25	<0.0001
Late Amniotomy	7:58	2:57	

Table 9: Table showing the time taken from induction to delivery in the study population.

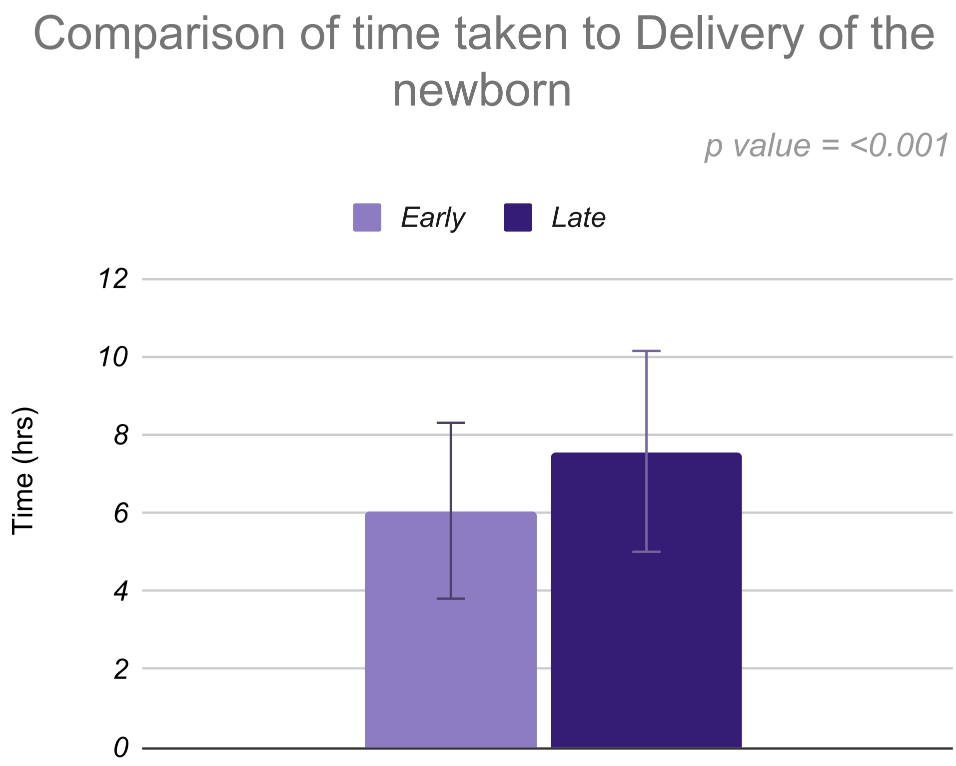

Figure 9: Chart showing the time taken from induction to delivery in the study population.

MODE OF DELIVERY	EARLY (n=70)	(%)	LATE (n=70)	(%)
Vaginal	44	62.86	31	44.29
Caesarean	24	34.29	38	54.29
Instrumental Delivery	2	2.86	1	1.43

Table 10: Table showing the distribution of women in different age groups in the study population.

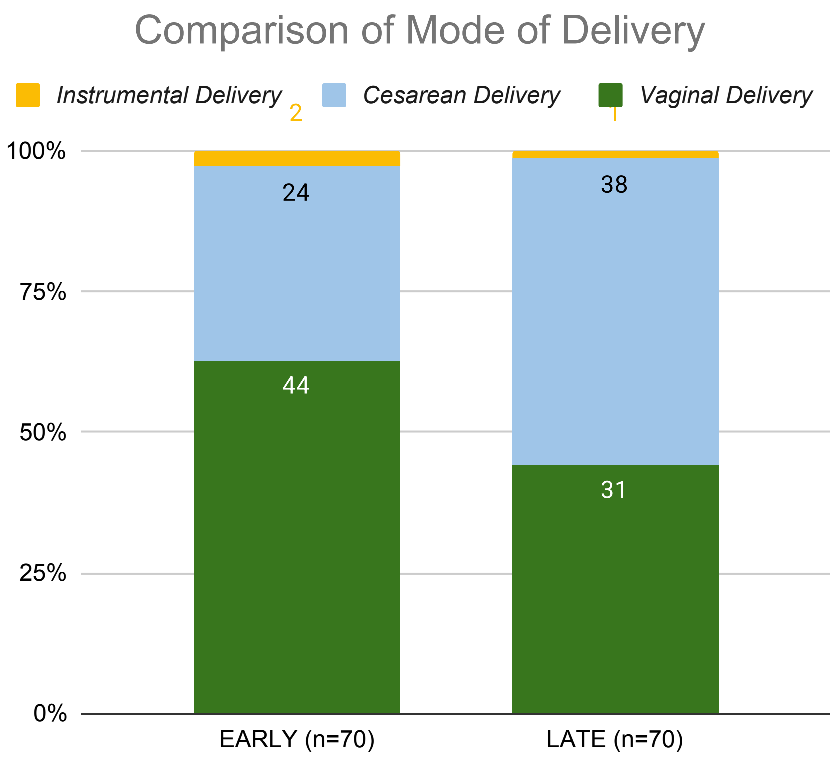

Figure 10: Chart showing the distribution of women in different age groups in the study population.

CESAREAN INDICATION	**EARLY (n=24)**	**(%)**	**LATE (n=38)**	**(%)**
Failed induction	6	25	15	39.47
Fetal distress	14	58.33	13	34.21
Arrest of labour	4	16.67	10	26.32

Table 11: Table showing the various indications for caesarean section the study population.

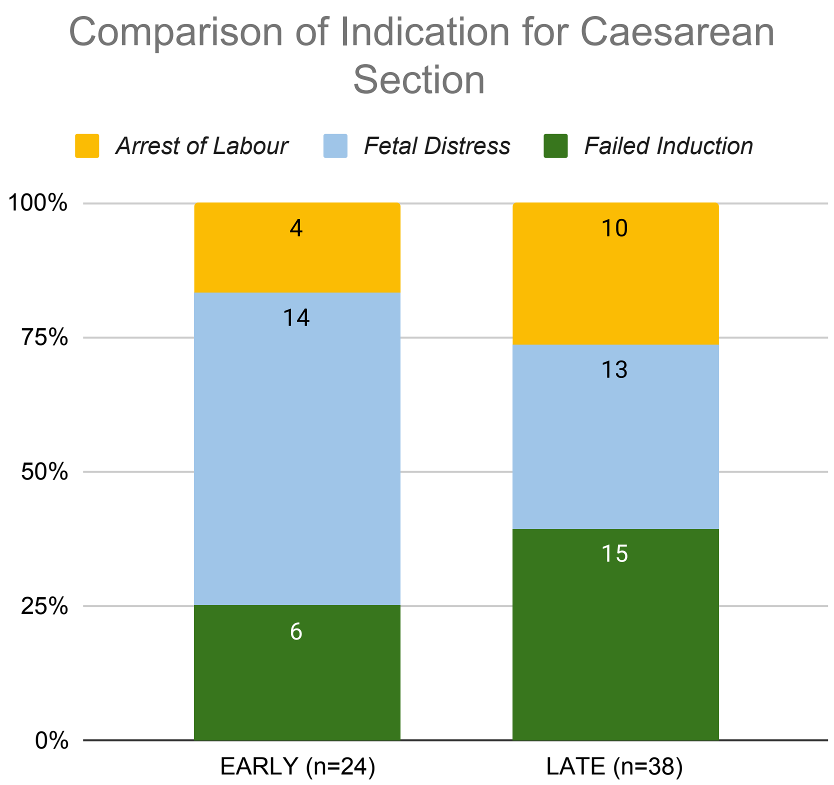

Figure 11: Chart showing the various indications for caesarean section the study population.

CTG ABNORMALITY	EARLY (n=70)	%	LATE (n=70)	%
YES	6	8.57	15	21.43
NO	64	91.43	55	78.57

Table 12: Table showing the incidence of Cardiotocographic abnormality in the study population.

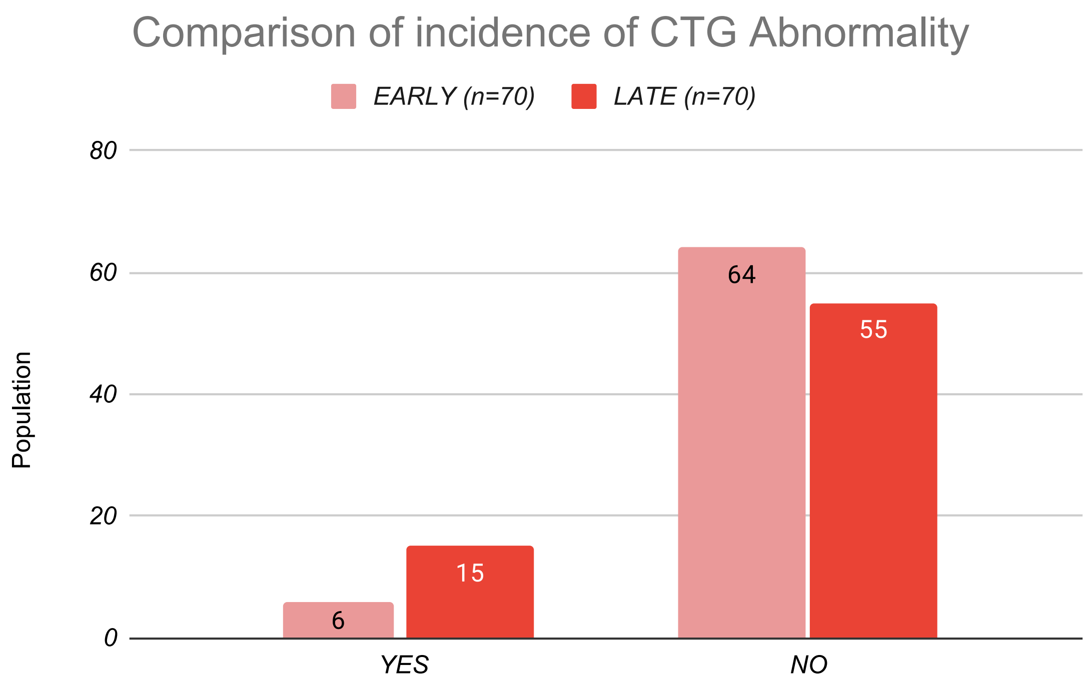

Figure 12: Chart showing the incidence of Cardiotocographic abnormality in the study population.

UTERINE HYPERSTIMULATION	**EARLY (n=70)**	**%**	**LATE (n=70)**	**%**
YES	1	1.43	0	0
NO	69	98.57	70	100

Table 13: Table showing the incidence of post uterine hyperstimulation in the study population.

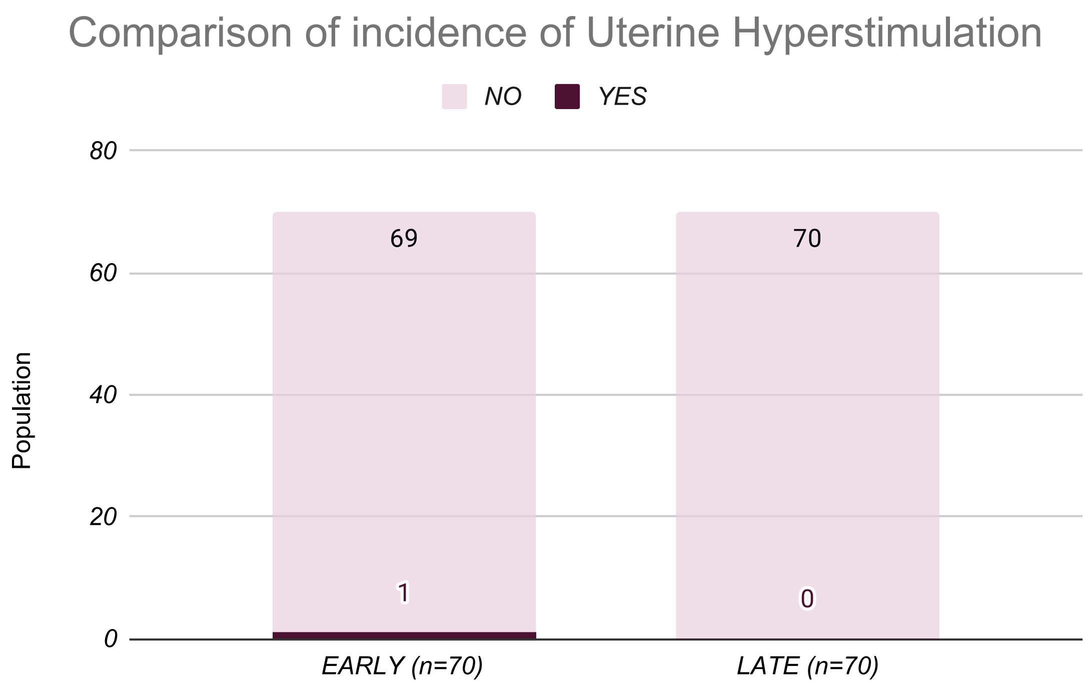

Figure 13: Chart showing the incidence of post uterine hyperstimulation in the study population.

Post Partum Hemorrhage	**EARLY (n=70)**	**%**	**LATE (n=70)**	**%**
YES	5	7.14	9	12.86
NO	65	92.86	61	87.14

Table 14: Table showing the incidence of post partum hemorrhage in the study population.

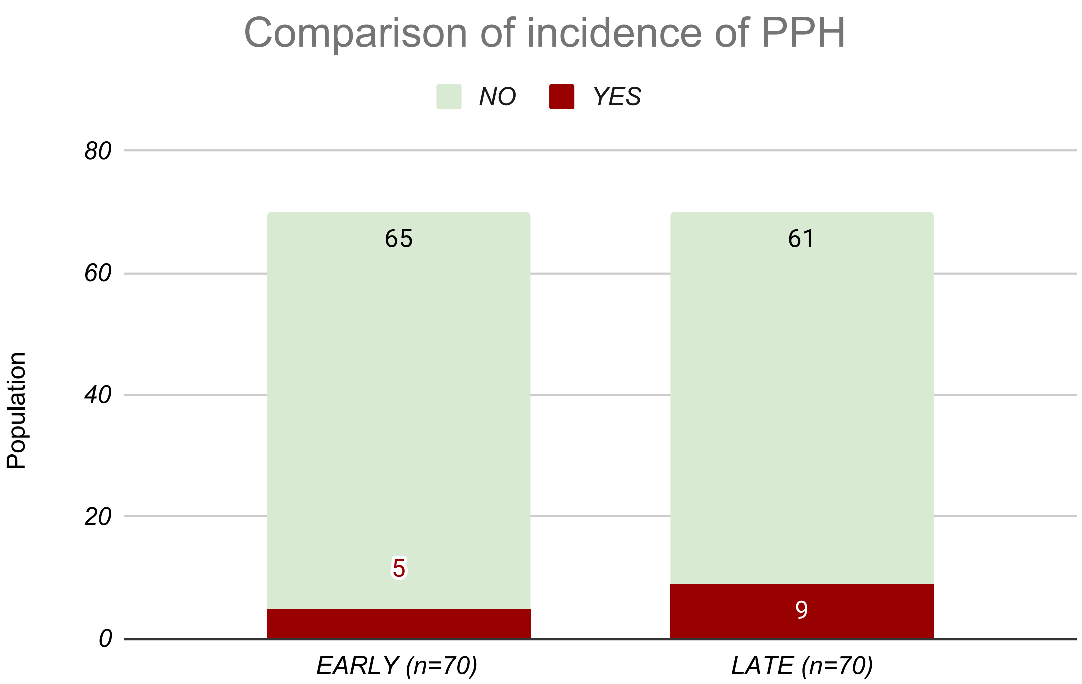

Figure 14: Chart showing the incidence of post partum hemorrhage in the study population.

	MEAN(mL)	STD DEV(mL)	p-value
Early Amniotomy	464.64	210.44	0.0352
Late Amniotomy	543.21	226.38	

Table 15: Table showing the the distribution of blood loss during delivery in the study population.

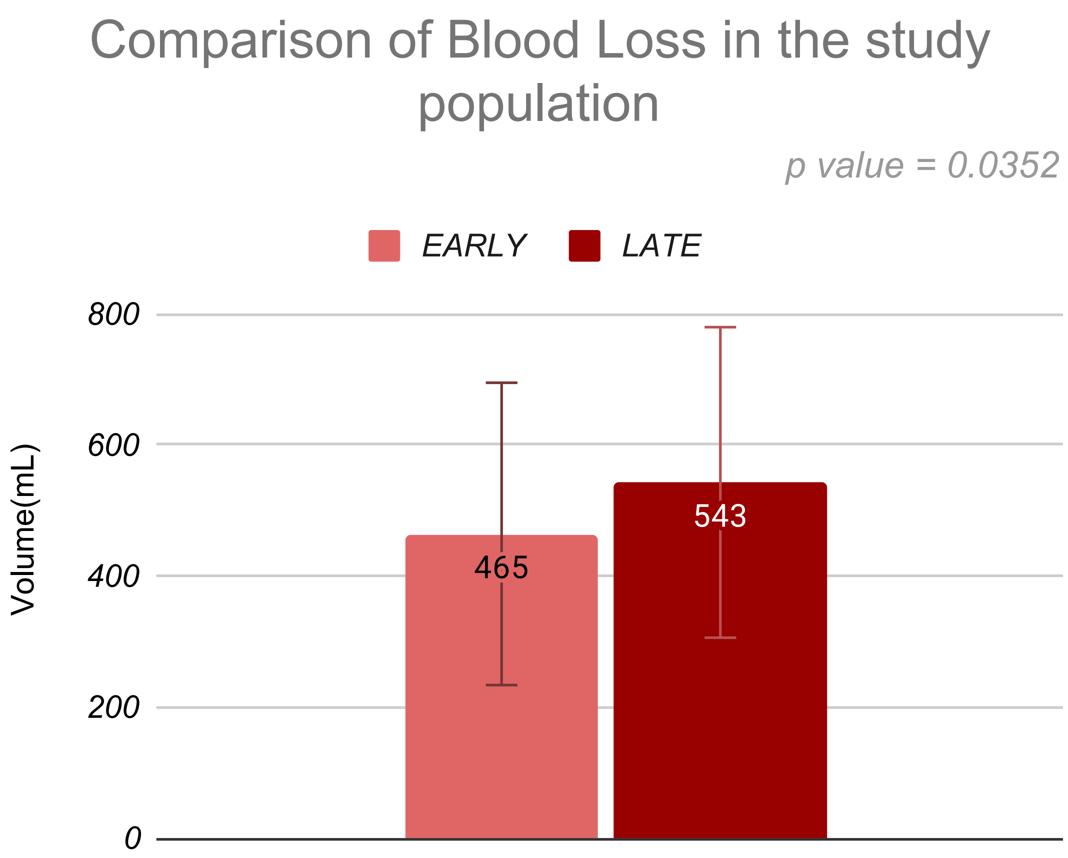

Figure 15: Chart showing the the distribution of blood loss during delivery in the study population.

	MEAN(mL)	STD DEV(mL)	p-value
Early Amniotomy	666.67	230.74	0.8114
Late Amniotomy	651.97	237.91	

Table 16: Table showing the the distribution of blood loss during delivery in women undergoing caesarean section in the study population.

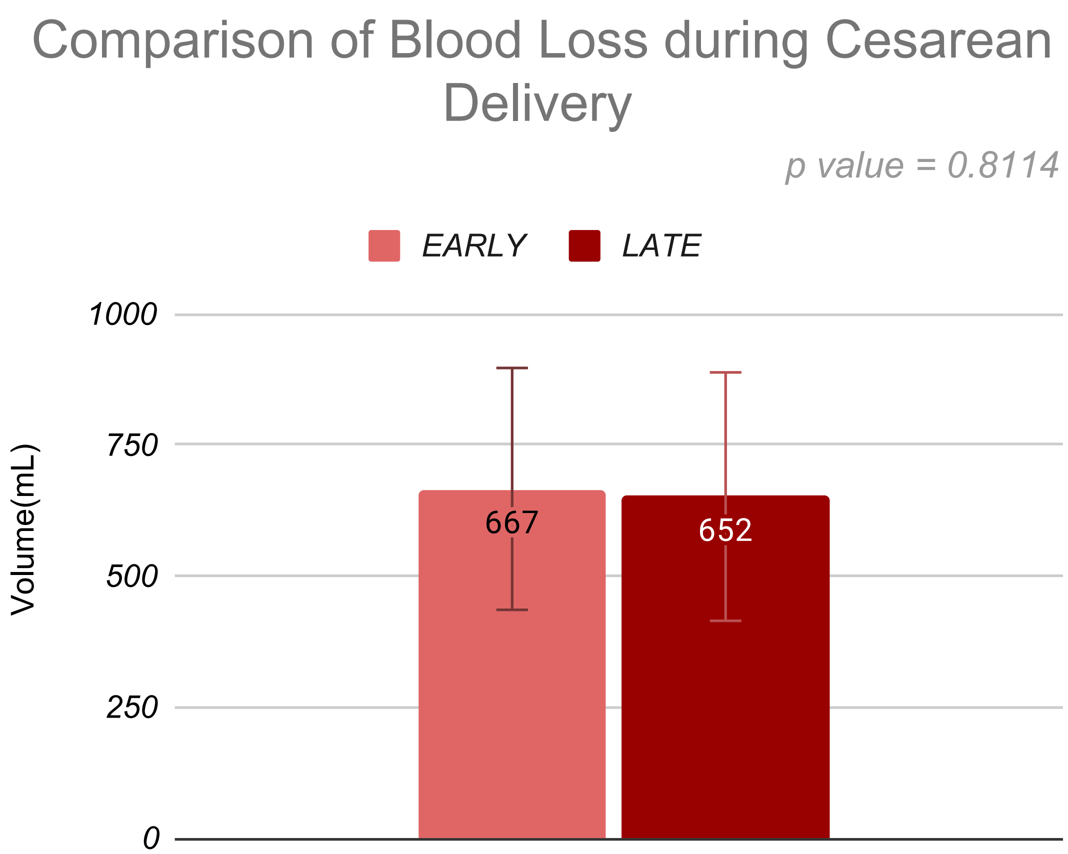

Figure 16: Chart showing the the distribution of blood loss during delivery in women undergoing caesarean section in the study population.

	MEAN(mL)	STD DEV(mL)	p-value
Early Amniotomy	359.24	86.98	0.0228
Late Amniotomy	414.06	121.64	

Table 17: Table showing the the distribution of blood loss during delivery in women undergoing normal vaginal delivery in the study population.

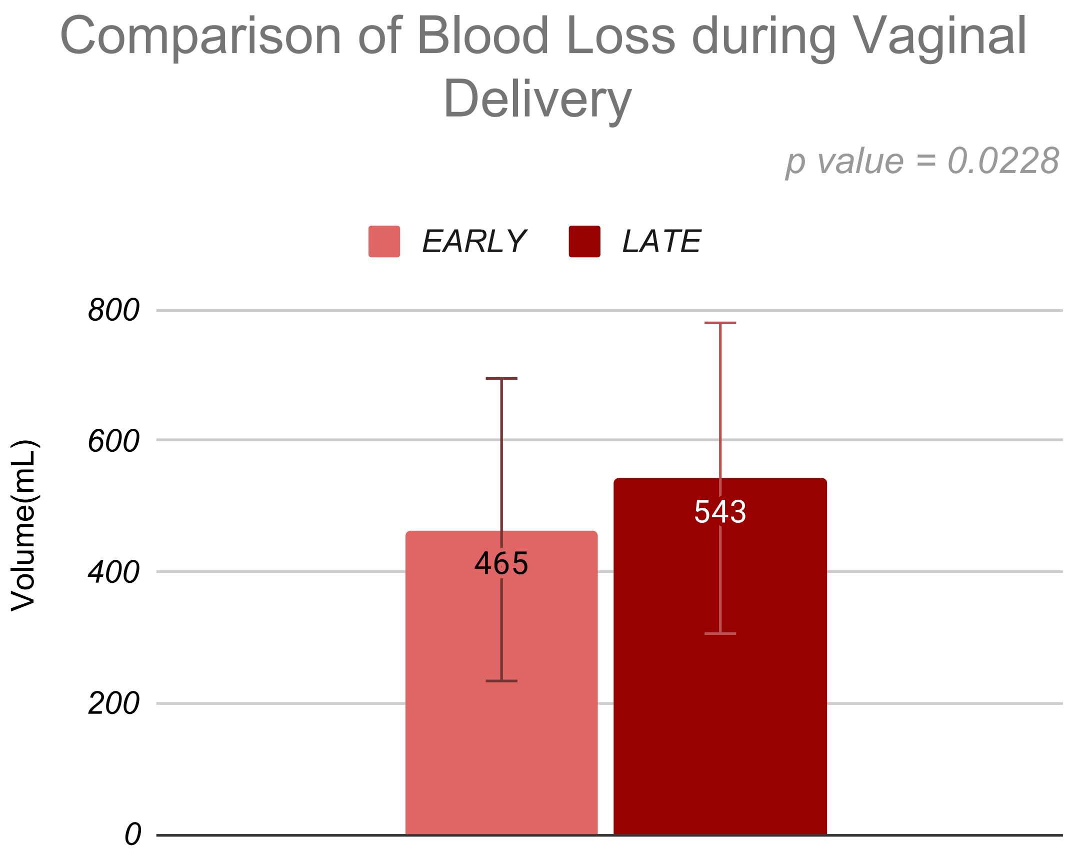

Figure 17: Chart showing the the distribution of blood loss during delivery in women undergoing normal vaginal delivery in the study population.

MATERNAL FEVER	**EARLY (n=70)**	%	**LATE (n=70)**	%
YES	1	1.43	2	2.86
NO	69	98.57	68	97.14

Table 18: Table showing the incidence of Maternal fever in the study population.

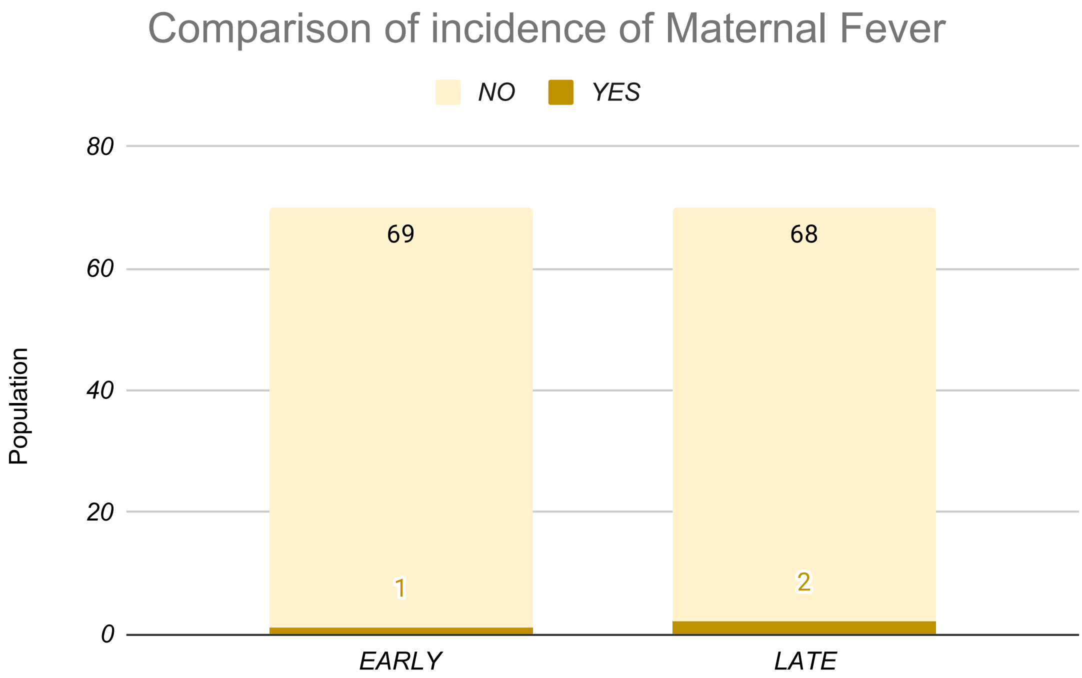

Figure 18: Chart showing the incidence of Maternal fever in the study population.

NICU ADMISSION	EARLY (n=70)	%	LATE (n=70)	%
YES	13	18.57	22	31.43
NO	57	81.43	48	68.57

Table 19: Table showing the incidence of NICU admission in the study population.

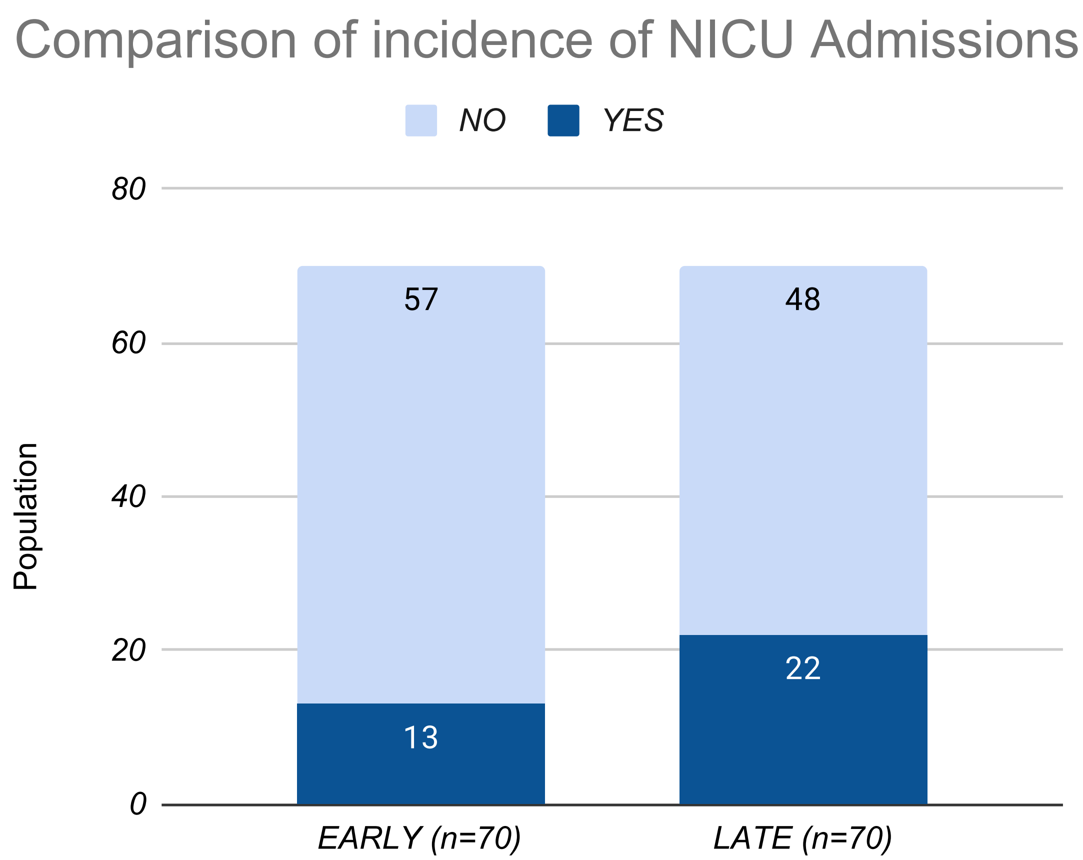

Figure 19: Chart showing the incidence of NICU admission in the study population.

	MEAN(g)	**STD DEV(g)**	**p-value**
Early Amniotomy	3017.34	294.58	0.1597
Late Amniotomy	2947.97	285.91	

Table 20: Table showing the the distribution of birth weight of the newborns in the study population.

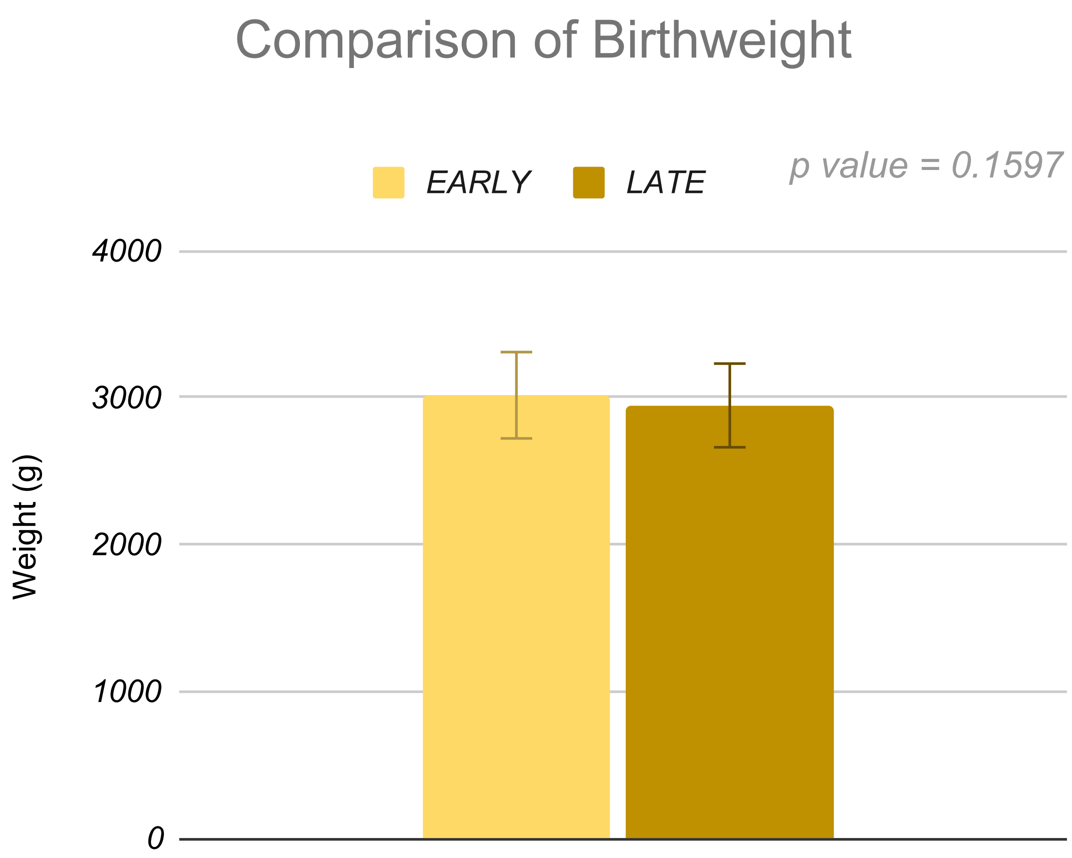

Figure 20: Chart showing the the distribution of birth weight of the newborns in the study population.

APGAR	EARLY (n=70)	(%)	LATE (n=70)	(%)
≤7	21	30	24	34.29
>7	49	70	46	65.71

Table 21: Table showing the APGAR scores in the study population.

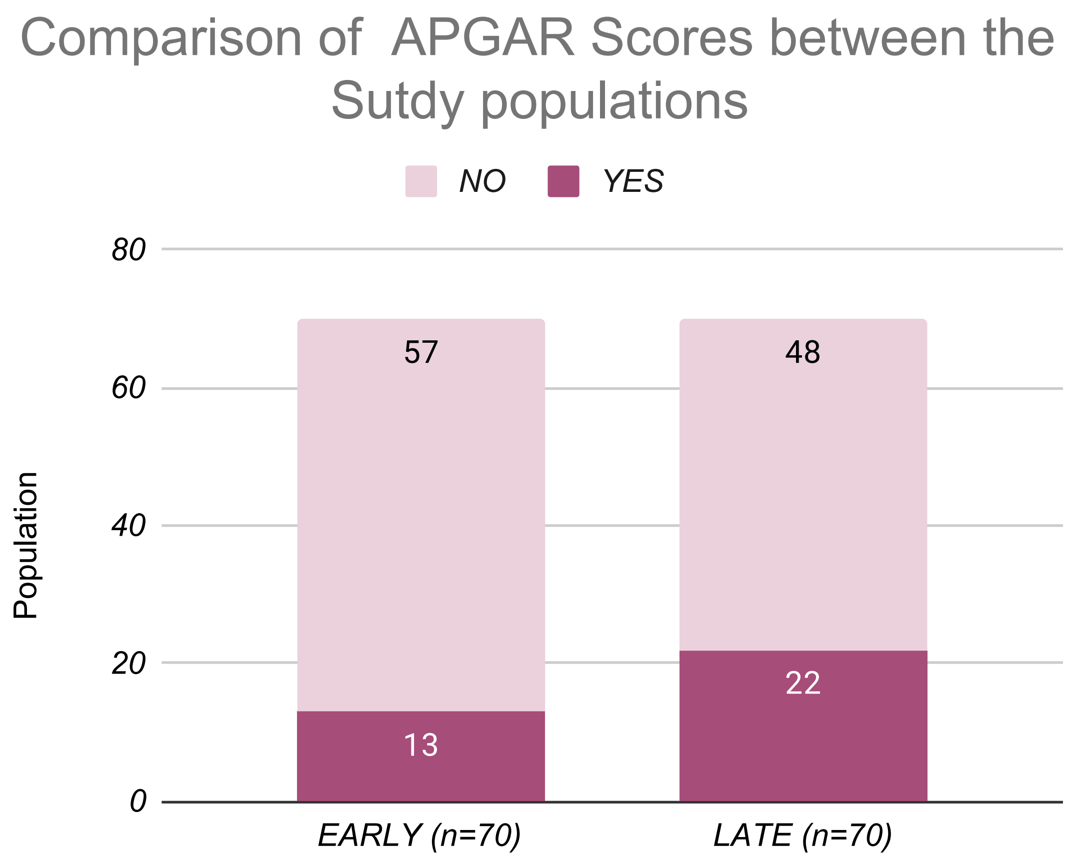

Figure 21: Chart showing the APGAR scores in the study population.

STILL BIRTH	**EARLY(n=70)**	**LATE(n=70)**
YES	0	0
NO	70	70

Table 22: Table showing the incidence of stillbirth in the study population.

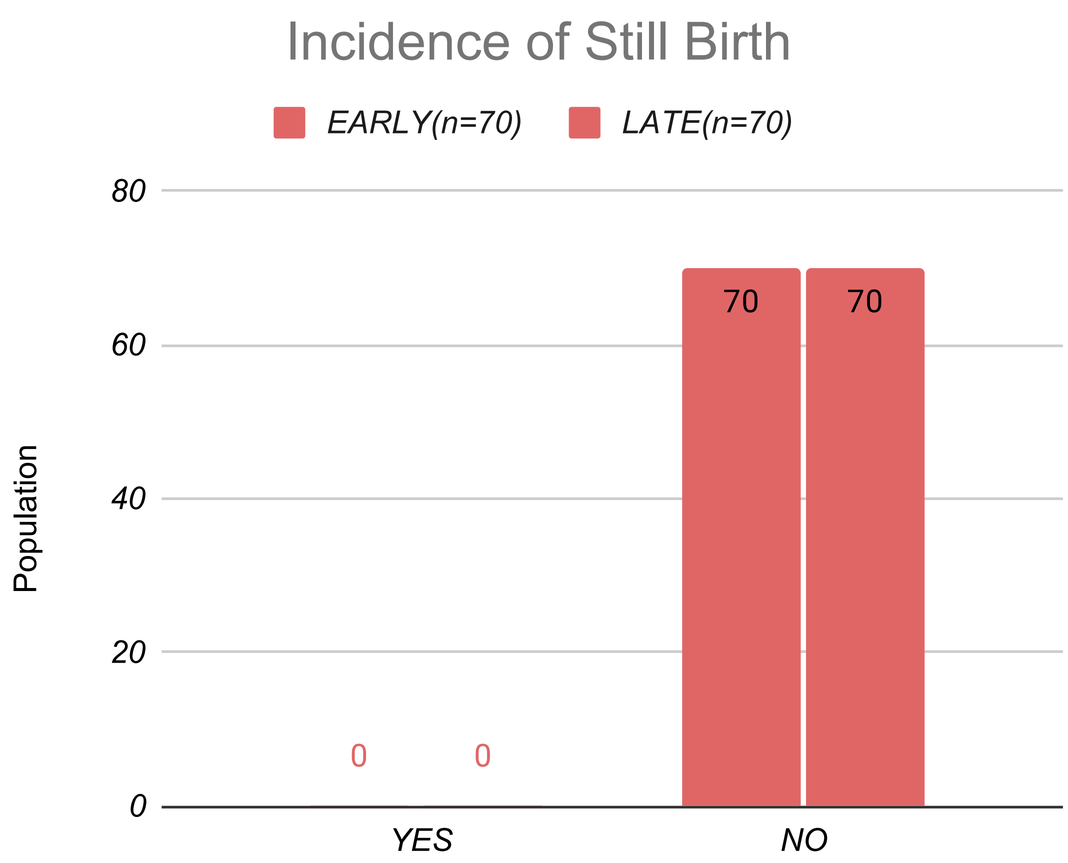

Figure 22: Chart showing the incidence of stillbirth in the study population.

DISCUSSION

Out of 150 patients who were included in out study 36% of the population were aged less than 21 years while the majority of around 57.33% belonged to the ages between 21 and 30. A small but significant population of 6.67% belonged to the age of 30 to 35years. 75% of the study population were booked before the bleeding episode. 25% of the study population were unbooked. These women were unbooked as pregnancy was diagnosed as a result of evaluation of the bleeding episode. 64% of the women experienced bleeding for the first time in the current pregnancy while 35.33% of the women had history of similar bleeding in the previous pregnancy.

60.67% of the study population were primigravida whereas 39.33% were multigravidas. Among multigravidas 1.33% were grand multigravidas. 14% of the study population had bleeding episode before 6 weeks of gestation while 20% had their first bleeding episode after 10 weeks. The majority of the population experience bleeding between 6 and 10 weeks. Most of the women experience mild spotting as described by them. These women constituted 48% of the study population. 28% of the women experience bleeding that they considered "more than spotting". 24% of the women had bleeding that had clots.

In our study 84% of the women with spotting in the first trimester continued their pregnancy beyond the viability period and hence were diagnosed as threatened abortion only. The number of women who continued their pregnancy dropped to 76.19% in the more than spotting group while it was only 27.78% in the women with clots. 6.94% of the spotting group went on to have complete abortion while 11.9% in the more than spotting group went on to have complete abortion. Interestingly none of the women in the clots group had complete abortion. 72.22% of the clots group were diagnosed with incomplete abortion. This pattern may be because women with complete abortion expelled the fetus completely with minimal bleed afterwards and hence the number of women who reported the bleeding as clots was negligible. Only 1.39% in the spotting group and 7.14% in the more than spotting group had incomplete abortion.

5.56% in the spotting group were diagnosed with ectopic pregnancy while there were none of the women in the more than spotting and clots group had ectopic pregnancy. A small number of women in the spotting and more than spotting group had molar pregnancies accounting for 1.39% 5and 4.7% respectively. None of the clots group were diagnosed with molar pregnancy.

Among the 61 pregnancies that progressed beyond the first trimester in the spotting only group 3.28% ended up in miscarriage. The rest 96.72% delivered a live neonate. 16.39% delivered a preterm baby.

80.33% delivered a term neonate.

32 women in the more than spotting group progressed beyond the first trimester. The number of miscarriages beyond first trimester doubled in the more than spotting group to 6.25%. The majority of women who delivered a live neonate delivered prematurely accounting for about 78.13% while term neonates constituted only 15.63%. Only 10 women who had clots in the first trimester progressed beyond 12 weeks. 2 women of that 10 miscarried soon after the first trimester. 6 women delivered a preterm neonate while 2 delivered a term neonate.

Among the 41 women who delivered prematurely the NICU admission rate was 75.61%. This rate was only 10.71% in the term neonates. 12.2% of the preterm neonates had a birthweight of less 1.5kg. 24.39% of preterm neonates weighed between 1.5 and 2.5 and 63.41% weighed between 2.6 and 3kg. None of the neonates who delivered prematurely weighed more than 3kgs. All the term neonates weighed above 1.5kg with more than 90% of neonate weighing above 2.5kg. 62.5% term neonates weighed between 2.5 and 3kg while 28.67% weighed more than 3kg.

The number of preterm neonates who had an APGAR of <7 was little less than half of the total. 43.9% preterm neonate had an APGAR of <7 and 56.10% neonates had an APGAR of >7. As expected 90% of term nenoates had an APGAR of >7 while the rest 10% had an APGAR of

<7. The low APGAR in these neonates may be due to the obstetrical complications seen in the mothers. 4.12% of mothers had placental abruption, 15.46% women had hypertensive disorders of pregnancy, 11.34% had gestational diabetes mellitus, 8.25% had both hypertension and diabetes and 9.28% had placenta previa. Placental abruption was only seen in women who delivered at term possibly due to placental abnormalities that progressed as the pregnancy prolonged. All other complications were similar in both the term and preterm population.

48.45% of the mothers delivered vaginally. 5.15% had instrumental delivery and 46.39% had caesarean delivery. Among the 97 neonates delivered 2(2.06%) were still born and there were 5 neonatal deaths (5.15%).

CONCLUSION

CONCLUSION

BIBLIOGRAPHY

ANNEXURES

CERTIFICATE OF APPROVAL

PLAGIARISM CERTIFICATE

PROFORMA

Particulars

- Name
- Age
- Address
- Phone no
- Religion
- Occupation
- Diet:Veg/Non Veg
- Education
- Date of admission
- Date of discharge
- Chief complaints

PAST History

- Seizure disorder
- Hypertension

- Diabetes mellitus
- Stroke
- Ischaemic heart disease
- Pregnancy induced hypertension

Family history

- Diabetes mellitus
- Hypertension
- Stroke
- Ischaemic heart disease
- Seizure disorder

Obstetric history

- Gravida/parity
- Gestational age at the time of presentation
- Gestational age at the time of delivery

Treatment history

- Antihypertensive medication
- Antiepileptic medication
- Anticoagulants

Others

General

- PR
- B.P
- RR
- TEMP
- WT
- HT
- BMI
- Pallor
- Icterus
- Cyanosis
- Edema

- Lymphadenopathy
- Clubbing

CNS examination

CVS examination

Respiratory system examination

GI examination

OUTCOME OF INDUCTION

- Complete abortion
- Surgical interference

INDUCTION ABORTION INTERVAL

BLEEDING DURATION

DRUG SIDE EFFECTS

- Nausea
- Vomiting
- Diarrhoea
- Dizziness
- Headache

- Abdominal pain
- Fever
- Chills

Investigations

- Hb
- PBS
- Hct
- Coagulation profile
- MCV
- TLC
- DLC
- Platelet count
- Blood urea
- Sr.Creatinine
- Urine protein

- S.Uric acid
- S.Bilirubin
- SGOT
- SGPT
- Sr.ALP
- Total Protein
- Sr.LDH
- Sr.Sodium
- Sr. Potassium

Ultrasound to confirm gestational age

PATIENT CONSENT FORM

Dissertation Topic: Comparison of Early Versus Late Amniotomy with Immediate Oxytocin Infusion In Nulliparous women with bishop's score $leq6$ - A Randomised Trial

Department: Dept. of Obstetrics and Gynaecology

Hospital Name: Govt. Raja Mirasudhar Hospital, Thanjavur Medical College

Patient's Name:

Patient's Age

Patient's Ip No:

Mark the following (x) : ( )The nature of the study explained to me clearly. I have been given the opportunity to clear my queries regarding the procedure

( ) I am willingly participating in this study. At any point of time I can get out of this study without any further issues

( ) During the study period or after the completion of this study, the doctor can verify my medical records without my consent

( ) I am willing to participate in this study and I am accepting this in my full conscience

Patients Name	Investigator's Sign
Patients Signature	M Saranya